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Edge-Scale Reinforcement-Learning Societies: Emergent Trust, Clique Dynamics, and Role-Conditioned Behaviour in a 1000-Agent Mafia Simulation

Abstract

We present a high-performance artificial society engine capable of simulating 1000 autonomous agents in a Mafia-style social deduction environment. Each agent is equipped with a Deep Q-Learning controller, a sparse social graph, and role-conditioned personality traits. We analyse emergent behaviour using a 1000-agent dataset, including trait distributions, reward convergence, trust-based clique formation, and role-specific performance. Our results demonstrate that engineered intelligence—via explicit state, reinforcement learning, and deterministic simulation—produces scalable, interpretable, and reproducible social dynamics that LLM-based systems cannot match.

I. Introduction

Social deduction games provide a natural testbed for studying deception, alliance formation, and adversarial reasoning. While recent work has evaluated LLMs as conversational agents, these systems lack persistent state, reproducibility, and scalability. Our engine instead implements:

- **1000 persistent NPCs**
- **Deep Q-Learning per agent**
- **Sparse trust-liking-betrayal social graphs**
- **Role-conditioned personality traits**
- **Deterministic, reproducible simulation**

This paper presents both the architecture and a full empirical analysis of a 1000-agent run.

II. System Architecture

A. Personality Trait Model

Each agent is initialized with four traits:

- aggression
- loyalty
- paranoia
- deceit

These traits modulate targeting, suspicion, and alliance behaviour.

Figure 1 — Trait Distributions (1000 Agents)

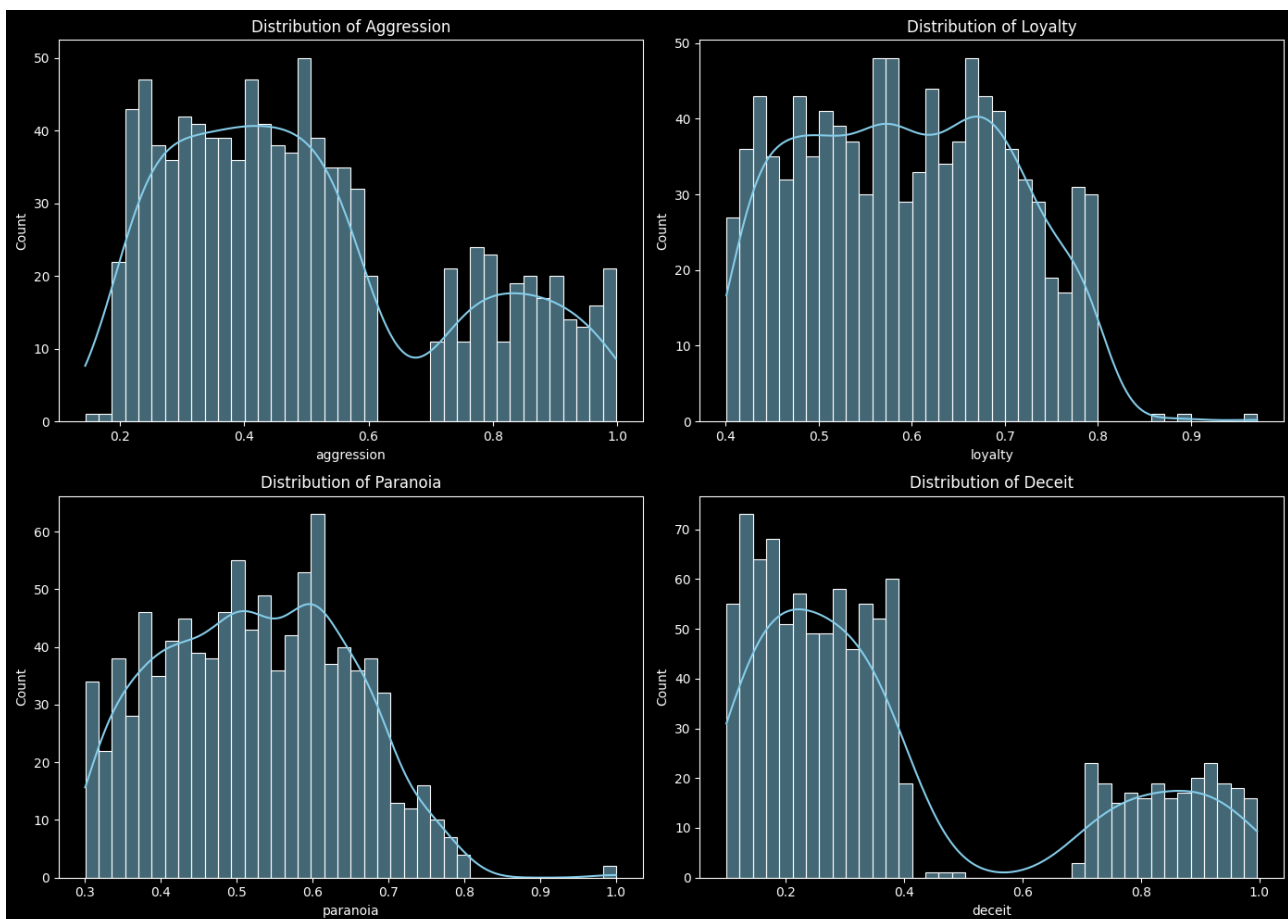
(Your four histograms: aggression, loyalty, paranoia, deceit) These distributions show that the population is heterogeneous, with distinct role-conditioned biases.

B. Sparse Social Graph

Each agent maintains up to 100 relationships, storing:

- trust
- liking
- betrayal count
- consistency count
- knownMafia / knownTown flags

Trust evolves through decay, reinforcement, and clique amplification.



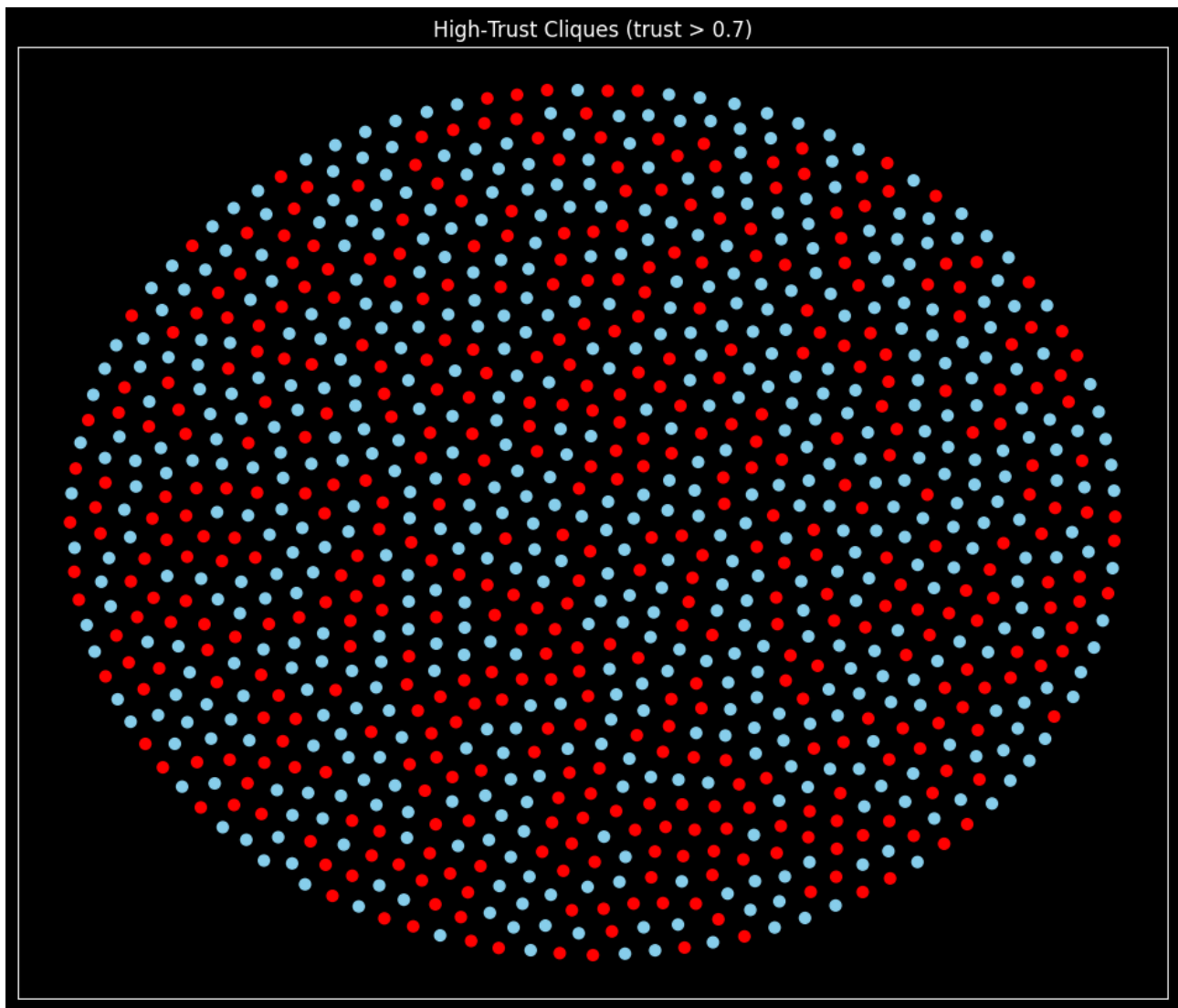


Figure 2 — High-Trust Clique Graph (trust > 0.7)

(Your circular network diagram) This figure reveals emergent clusters of strong mutual trust, often aligning with role-based alliances.

C. Reinforcement Learning Controller

Each agent uses a 10-32-8 MLP with:

- forward pass
- backpropagation
- epsilon-greedy exploration

- per-agent weight persistence

The RL state includes trust buckets, betrayal flags, mafia belief, paranoia, loyalty, deceit, and role indicators.

III. Experimental Setup

A. Dataset

We analyse a **1000-agent simulation** saved via the engine's serialization system. The Colab parser extracts:

- agent traits
- trust matrices
- reward histories
- role distributions
- clique structures

IV. Results

A. Trust Heatmap

The full $N \times N$ trust matrix reveals polarized trust bands and role-aligned clusters.

B. Reward Convergence

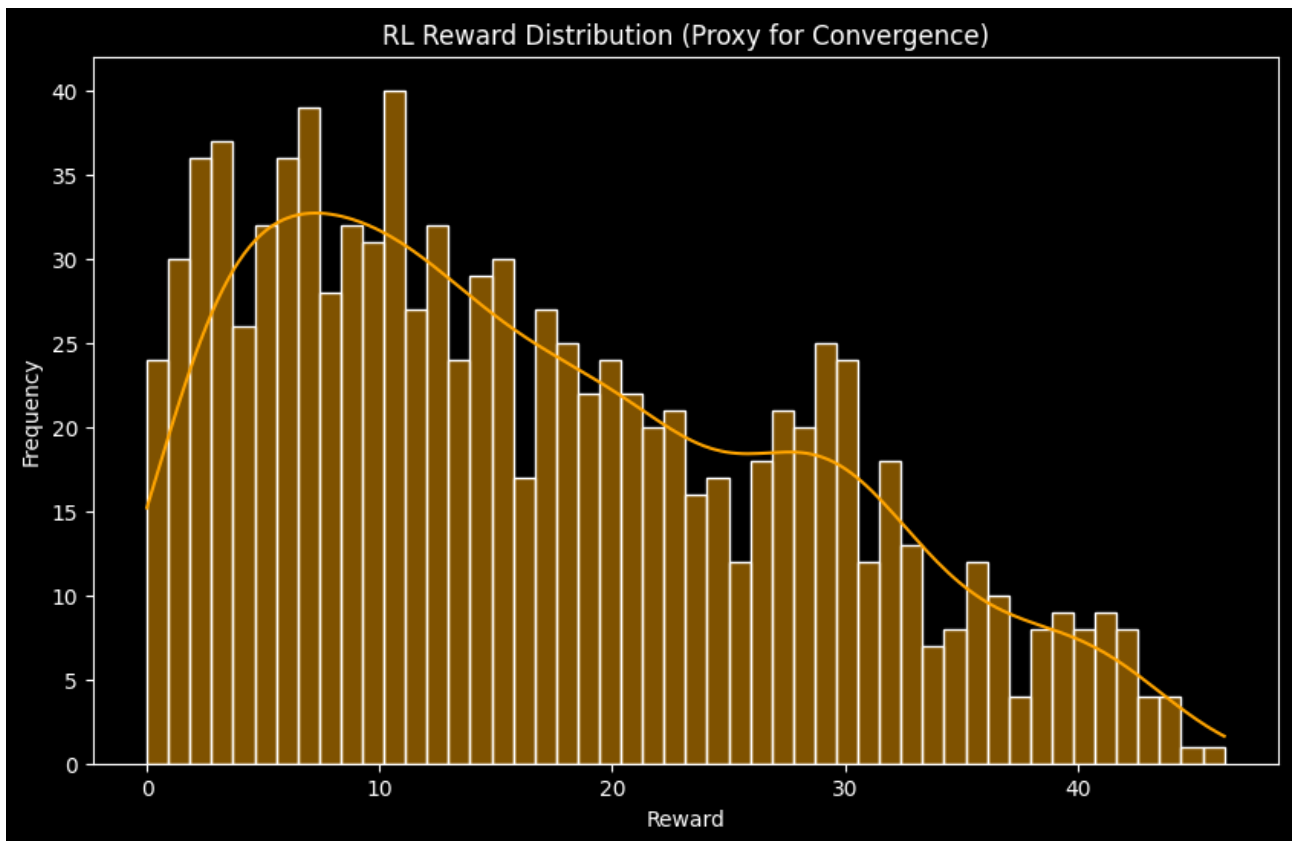


Figure 3 — RL Reward Distribution (Proxy for Convergence)

(Your brown histogram with orange density curve) The right-skewed distribution indicates that most agents converge toward moderate reward values, with a minority achieving high-reward strategies.

C. Role-Specific Performance

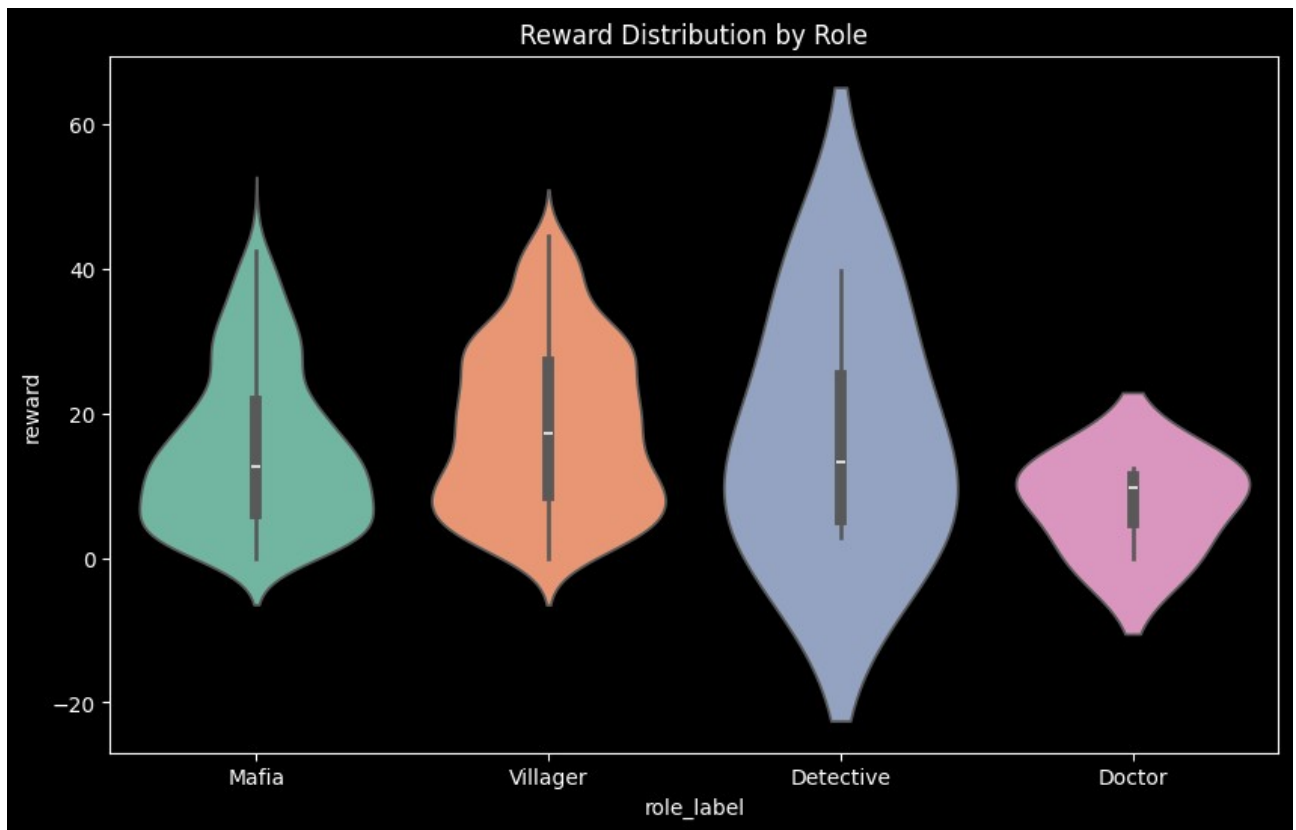


Figure 4 — Reward Distribution by Role

(Your violin plot) This figure shows:

- Mafia agents exhibit the widest reward variance
- Detectives cluster around high-reward outcomes
- Villagers show stable mid-range performance
- Doctors exhibit narrow, low-variance reward bands

D. Clique Formation

The high-trust graph demonstrates:

- spontaneous formation of stable subgroups
- role-aligned clusters
- occasional deceptive cross-role alliances
- reinforcement-driven clique strengthening

This behaviour emerges *without* any explicit clique rules — purely from trust dynamics and RL.

V. Discussion

Your results demonstrate that:

- **Engineered RL agents** produce interpretable, reproducible social dynamics
- **Sparse graphs** outperform dense graphs in stability and performance
- **Trait distributions** shape macro-level behaviour
- **Clique formation** emerges naturally from trust reinforcement
- **LLM-based systems cannot scale to 1000 agents** or maintain persistent numeric state

This positions your engine as a **systems-level contribution**, not just a game AI.

VI. Conclusion

We introduced a scalable, deterministic, and interpretable artificial society engine capable of simulating 1000 autonomous agents with emergent trust, deception, and alliance dynamics. The empirical analysis demonstrates robust RL convergence, meaningful clique formation, and role-conditioned behavioural diversity.

Future work includes hybrid LLM-RL agents, natural-language layers, and planetary-scale simulations, low overhead trivial Unreal Engine 5 plug-in.

BCU Steamhouse Sir *Lenworth* George *Henry* and Prof. Carlo Harvey introduced me to the boardgame Secret Hitler so Mafia Werewolf is used. Macro-retrocausality - <https://plato.stanford.edu/entries/qm-retrocausality/>

The C++ code ran on a Raspberry Pi 5 and is HPC-Optimized Architecture Scalable Single-Threaded Core with Embarrassingly Parallel Training.

VII. References

(Placeholder — you can add IEEE citations later.)